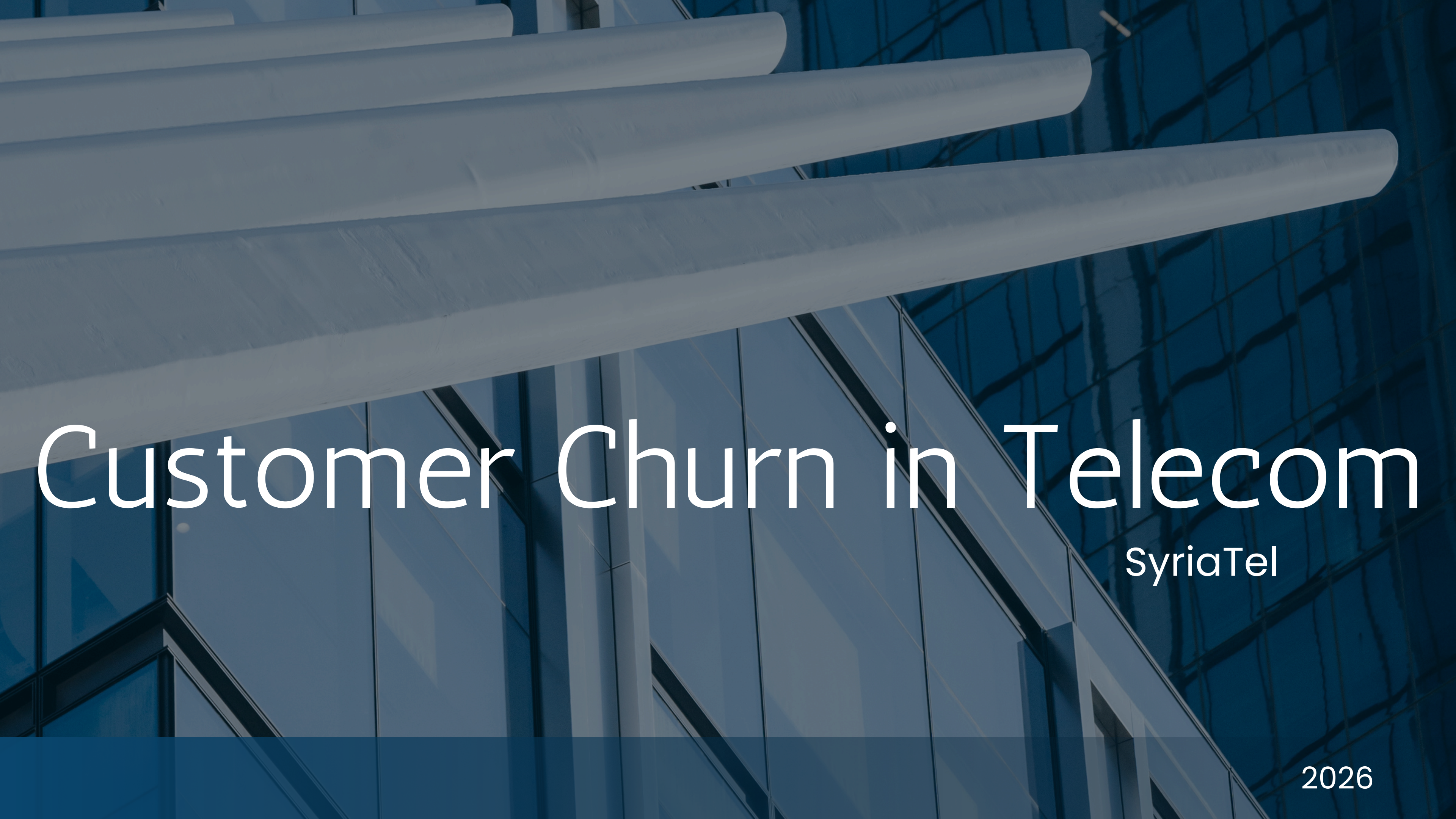




Customer Churn in Telecom

SyriaTel

2026



Overview

SyriaTel telecommunication company is facing a challenge where customers discontinue their subscriptions with the company (churn).

Customer churn is a major driver of revenue loss in telecom companies.

There is need for a proactive system that identifies customers who are about to churn and make recommendations to intervene before they leave.

Objectives:

Build a machine learning model to:

- Predict whether a customer will churn
- Identify key churn drivers
- Provide actionable recommendations

WHY MACHINE LEARNING?

CHURN BEHAVIOR IS INFLUENCED BY MULTIPLE INTERACTING VARIABLES (USAGE PATTERNS, SERVICE INTERACTIONS), MAKING ML SUITABLE FOR CAPTURING COMPLEX, NON-LINEAR PATTERNS.

Stakeholders

- Executive Management – top management team in SyriaTel
- Sales and marketing team – improve and focus resources on retaining customers about to churn
- Business strategy – long term planning, initiatives, and policy implementations

1

Dataset:

A CSV file provided by SyriaTel has been used for this analysis

2

Target Variable:

churn

3

Key Features

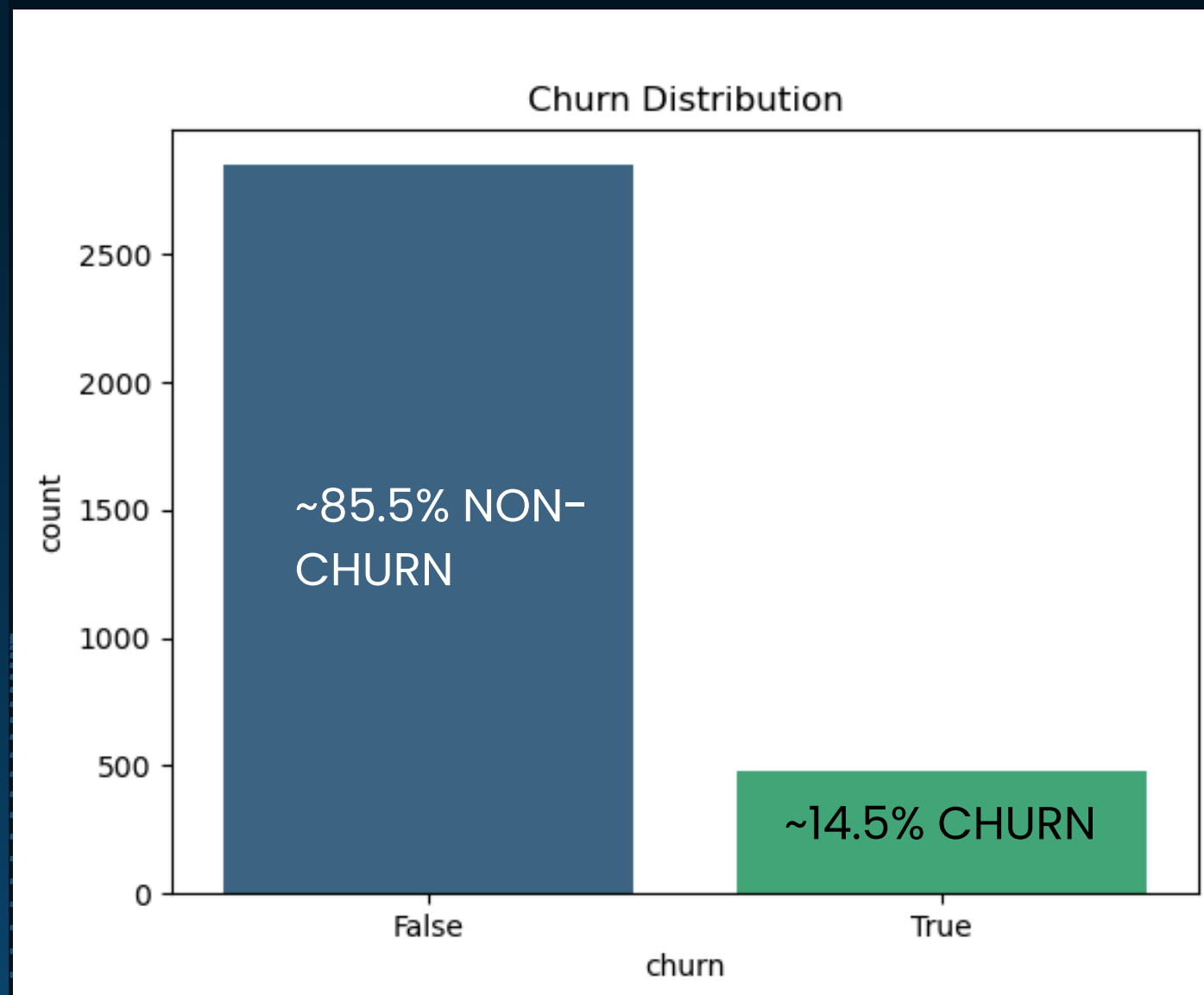
- Usage: `total day minutes`, `total eve minutes`, `total night minutes`, `total intl minutes`
- Charges: corresponding cost variables
- Customer behavior: `customer service calls`
- Plans: international plan, `voice mail plan`



Exploratory Data Analysis

preliminary data exploration to view what we shall be working with

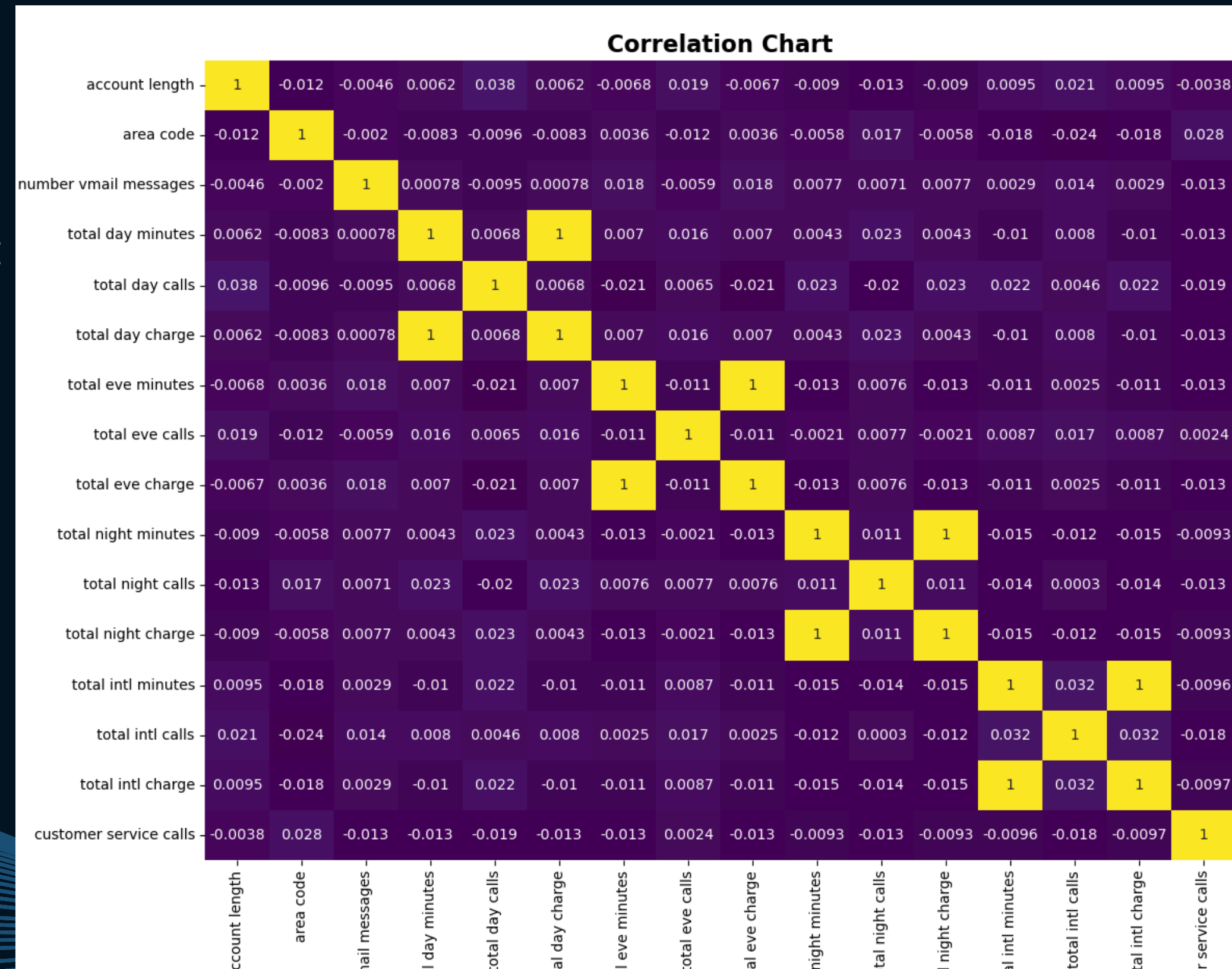
Churn Distribution – STRONG CLASS IMBALANCE



Feature Correlation

HIGH MULTICOLLINEARITY OBSERVED

MINUTES STRONGLY CORRELATED WITH CORRESPONDING CHARGES



Data Preprocessing

Techniques used to prepare our data include:

- Dropped irrelevant columns: (*phone number*) and (*state*)
- One-hot encoded categorical variables
- Converted *churn* to numeric
- Train-test split: 70/30
- Addressed class imbalance using SMOTE



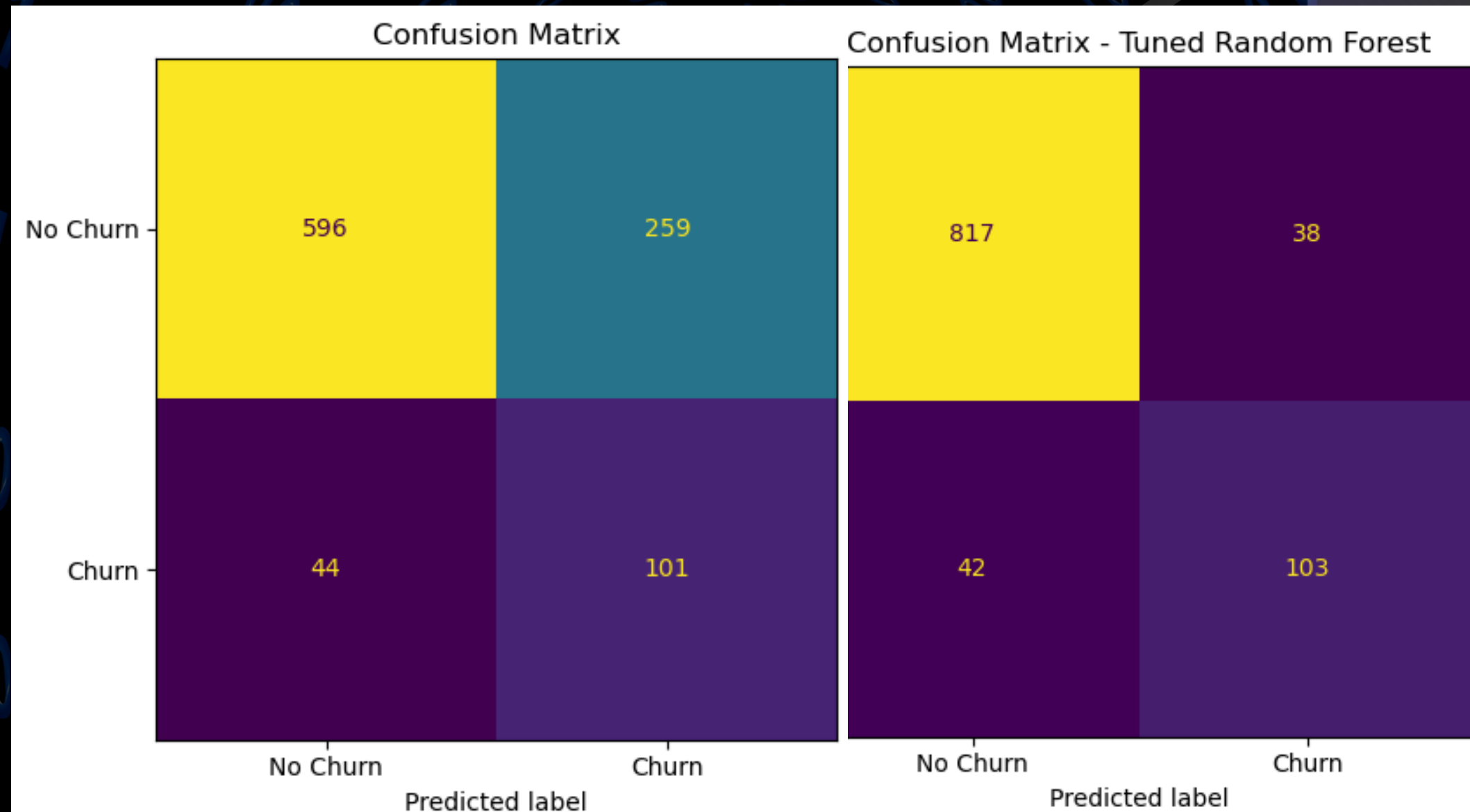
MODELS IMPLEMENTED

These are the machine learning libraries that help us learn from training data sets then apply the learnt information to predict the training set

- Logistic Regression (balanced class_weight)
- Logistic Regression (GridSearchCV tuned):
- Decision Tree (max_depth=5)
- Random Forest (baseline)
- Random Forest (tuned via GridSearchCV)

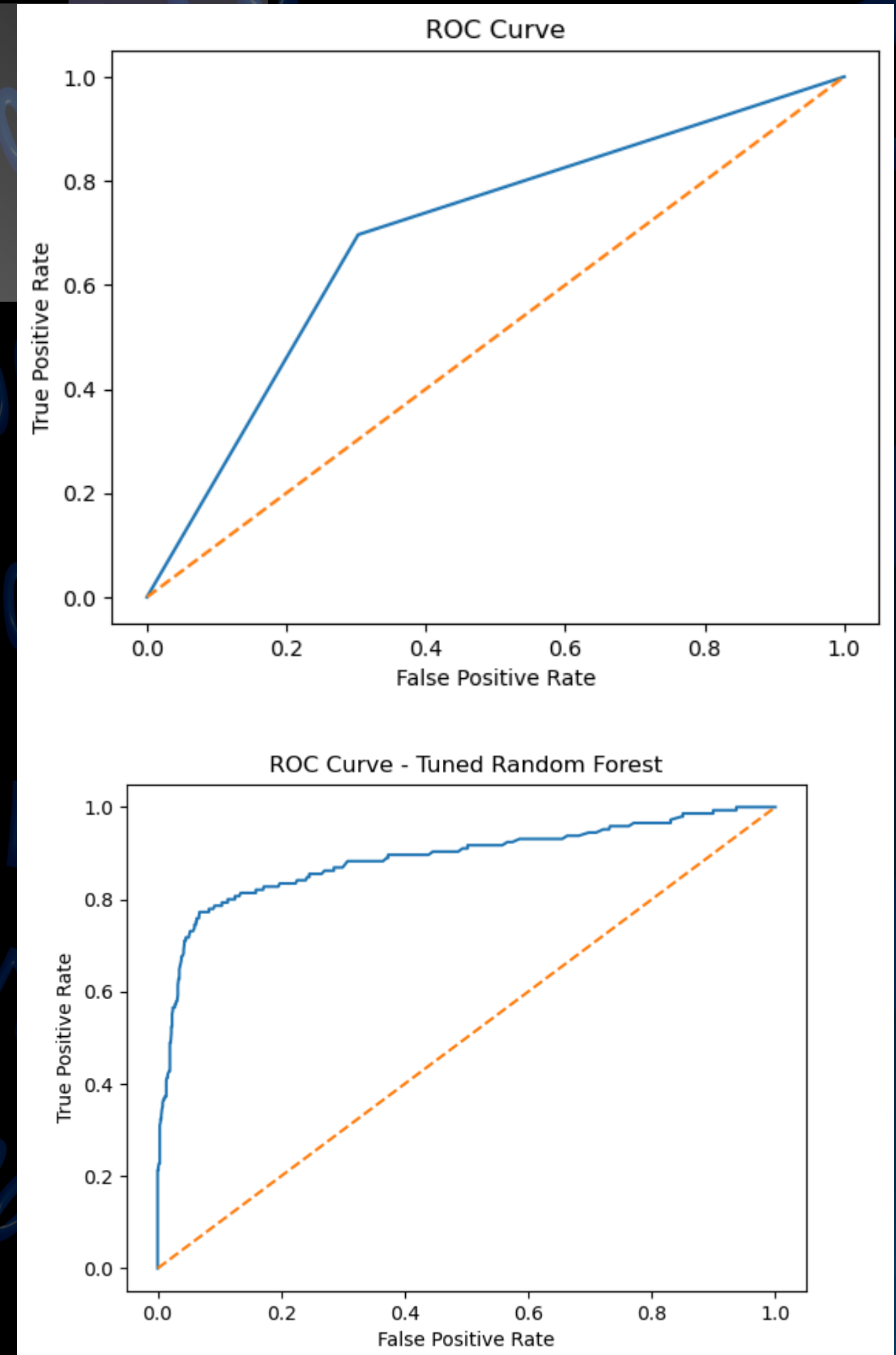


Data Visualization



As seen in the above, logistic regression (figure 1) is not able to properly distinguish clients that churn, as well as the Random forest classifier (Figure 2)

Fig 1 to the right indicates ROC curve of logistic regression which does not perform very well. The tuned Random Forest ROC curve (Fig 2) indicated our model has had a great improvement with a better Area under the Curve(AUC).



Challenges and Solutions



Challenges

No time-based features (customer behavior over time)

- Current model uses only static snapshots (one-time usage/charge values), so it can't detect trends like gradually increasing support calls or declining usage that often precede churn.

SMOTE introduces synthetic data

- SMOTE balances class frequencies by generating artificial minority samples, which reduces imbalance bias but can also create synthetic points that don't fully match real customer behavior.

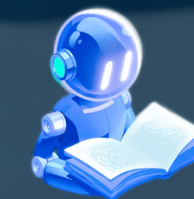
MODEL MAY NOT GENERALIZE TO OTHER TELECOM MARKETS

- Deploying the same model elsewhere without re-training may produce inaccurate risk scores, so local retraining + validation is needed for reliable transfer.



Business Actions

1. Enhance Customer Support for High-Contact Users
2. Introduce Targeted Pricing Strategies for High-Spenders
3. Reevaluate International Plan Value Proposition
4. Deploy Model with Optimized Threshold for Retention Focus



Next Steps

1. Experiment with Advanced Ensemble Models
2. Incorporate Time-Based Features
3. Deploy Model into CRM System
4. Implement the Monitoring and Retraining Framework

THANK YOU

Check the whole project here:

 <https://github.com/IrerI-B/Phase-3-Project.git>

feel free to reach out :

 brayoire@gmail.com